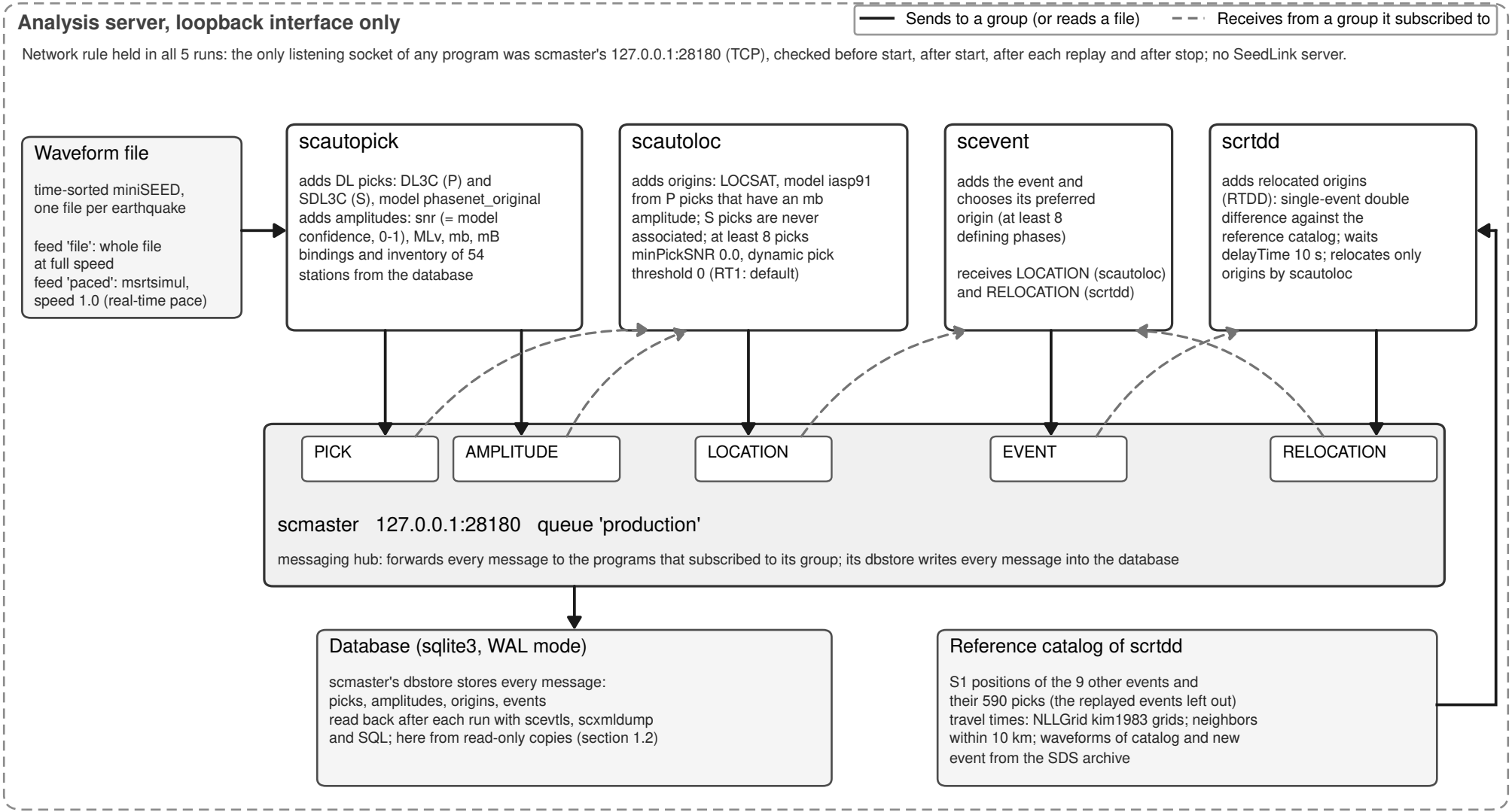
Analysis server, loopback interface only
Sends to a group (or reads a file)
Receives from a group it subscribed to
Network rule held in all 5 runs: the only listening socket of any program was scmaster's 127.0.0.1:28180 (TCP), checked before start, after start, after each replay and after stop; no SeedLink server.
Waveform file
time-sorted miniSEED, one file per earthquake
feed 'file': whole file at full speed
feed 'paced': msrtsimul, speed 1.0 (real-time pace)
scautopick
adds DL picks: DL3C (P) and SDL3C (S), model phasenet_original
adds amplitudes: snr (= model confidence, 0-1), MLv, mb, mB
bindings and inventory of 54 stations from the database
scautoloc
adds origins: LOCSAT, model iasp91
from P picks that have an mb amplitude; S picks are never associated; at least 8 picks
minPickSNR 0.0, dynamic pick threshold 0 (RT1: default)
scevent
adds the event and chooses its preferred origin (at least 8 defining phases)
receives LOCATION (scautoloc) and RELOCATION (scrtdd)
scrtdd
adds relocated origins (RTDD): single-event double difference against the reference catalog; waits delayTime 10 s; relocates only origins by scautoloc
PICK
AMPLITUDE
LOCATION
EVENT
RELOCATION
scmaster 127.0.0.1:28180 queue 'production'
messaging hub: forwards every message to the programs that subscribed to its group; its dbstore writes every message into the database
Database (sqlite3, WAL mode)
scmaster's dbstore stores every message: picks, amplitudes, origins, events
read back after each run with scevtls, scxmldump and SQL; here from read-only copies (section 1.2)
Reference catalog of scrtdd
S1 positions of the 9 other events and their 590 picks (the replayed events left out)
travel times: NLLGrid kim1983 grids; neighbors within 10 km; waveforms of catalog and new event from the SDS archive